Capstone Checkpoint 7.1 - BASELINE

Oracle, vector, and hybrid experiment comparison. Generated 2026-10-04 08:08 EDT.

6	2	114	19	0
Included runs	Configurations	Question-runs	Questions	Legacy runs excluded

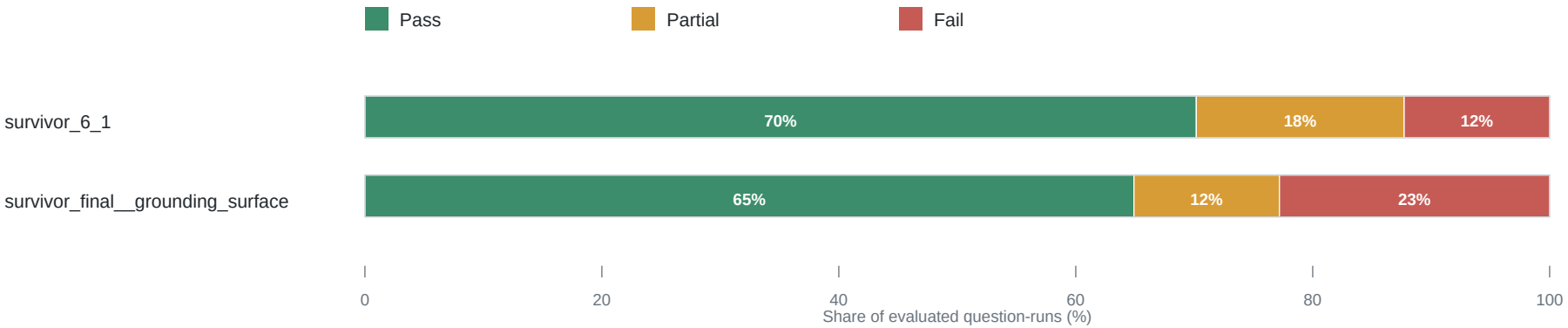
Summary

The highest observed mean quality score was 0.789 for survivor_6_1 across 3 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 100.0% for survivor_6_1. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 50.0 for Exhaustive list or time filter under survivor_6_1. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

Answer outcomes



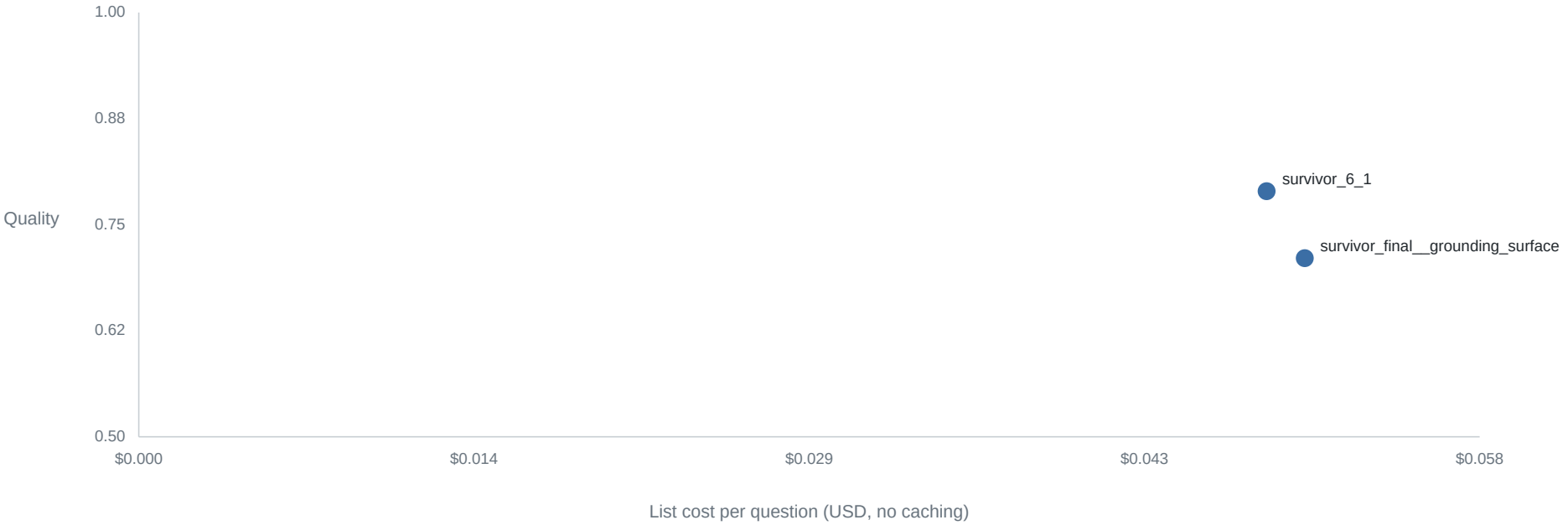
Configuration	Runs	Cases	Pass	Partial	Fail	Quality	Run SD	Article recall	Full source coverage	Mean context
survivor_6_1	3	57	70.2%	17.5%	12.3%	0.789	0.037	100.0%	100.0%	78.9k
survivor_final__grounding_surface	3	57	64.9%	12.3%	22.8%	0.711	0.043	100.0%	100.0%	86.7k

Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.

Cost and latency

Configurations whose runs carry per-question telemetry (the Checkpoint 6.1 cost starters). Every figure, quality included, is computed over those runs only, so cost and quality come from the same question-runs.

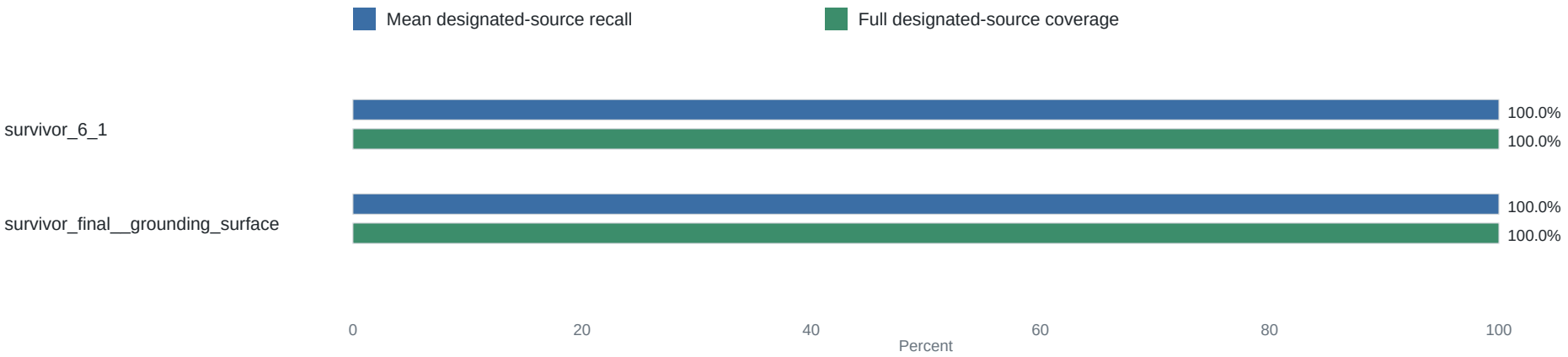
Configuration	Runs	Quality	Calls / question	Input tokens / question	Cached	List $ / question	Billed $ / question	Seconds / question	List $ per quality point
survivor_6_1	3	0.789	6.77	56.8k	35.2%	$0.0485	$0.0350	13.1	$0.0615
survivor_final__grounding_surface	3	0.711	6.84	59.3k	38.4%	$0.0502	$0.0348	13.8	$0.0706



List cost charges every input token at the full rate; billed cost is what OpenRouter reported, with cached input at the cached rate. The judge is not included. Seconds cover the agent graph and the answer call, not the judge.

Retrieval diagnostics

Designated-source coverage



The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.

Outcome and source-coverage combinations

Configuration	Pass, full	Pass, incomplete	Non-pass, full	Non-pass, missing
survivor_6_1	40	0	17	0
survivor_final__grounding_surface	37	0	20	0

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

	survivor_6_1	survivor_final__grounding_s..
GT01	**1.00**	**1.00**
GT02	**1.00**	**1.00**
GT03	**0.00**	**0.00**
GT04	**1.00**	**1.00**
GT05	**1.00**	**0.67**
GT06	**1.00**	**1.00**
GT07	**0.50**	**0.17**
GT08	**0.33**	**0.33**
GT09	**0.50**	**1.00**
GT10	**1.00**	**0.67**
GT11	**0.00**	**0.17**
GT12	**1.00**	**1.00**
GT13	**1.00**	**1.00**
GT14	**1.00**	**1.00**
GT15	**1.00**	**0.67**
GT16	**0.83**	**0.17**
GT17	**1.00**	**1.00**
GT18	**1.00**	**1.00**
GT19	**0.83**	**0.67**

Fail (0) Partial (0.5) Pass (1)

Question	survivor_6_1	survivor_final__grounding_surface
GT01	1.000	1.000
GT02	1.000	1.000
GT03	0.000	0.000
GT04	1.000	1.000
GT05	1.000	0.667
GT06	1.000	1.000
GT07	0.500	0.167
GT08	0.333	0.333
GT09	0.500	1.000
GT10	1.000	0.667
GT11	0.000	0.167
GT12	1.000	1.000
GT13	1.000	1.000
GT14	1.000	1.000
GT15	1.000	0.667
GT16	0.833	0.167
GT17	1.000	1.000
GT18	1.000	1.000
GT19	0.833	0.667

Per-question article recall

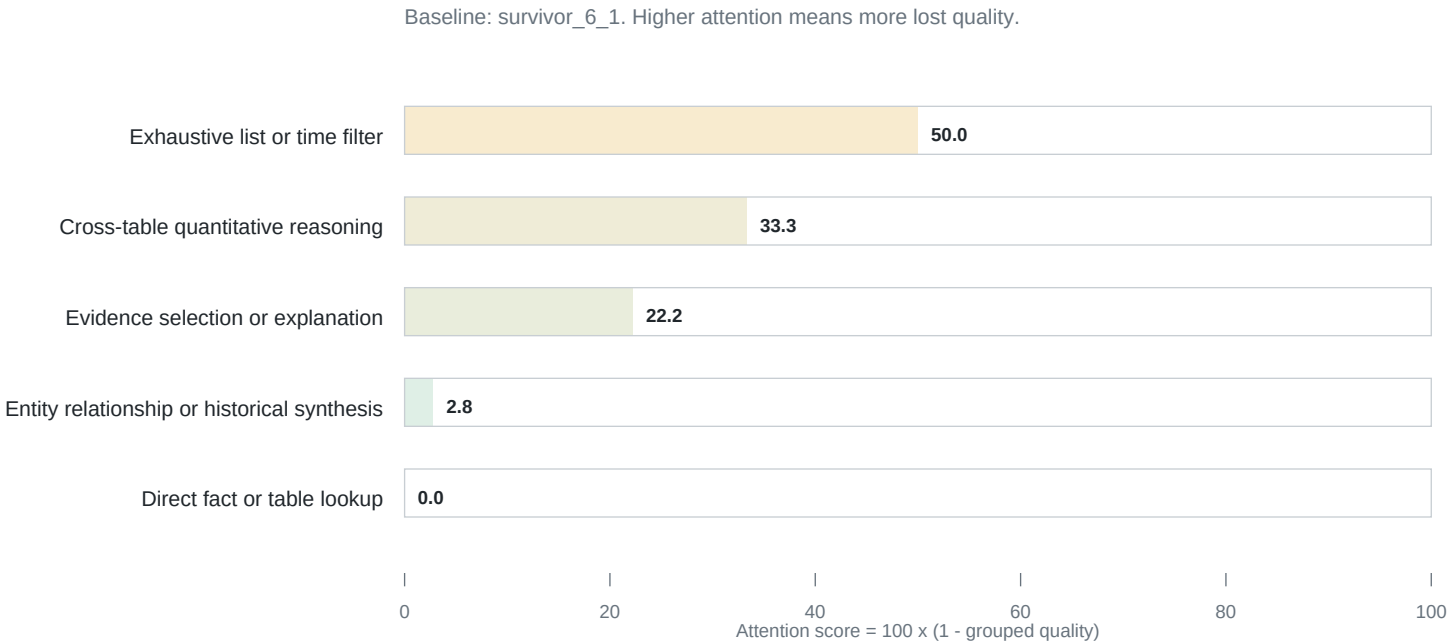
Fraction of each question's designated source articles that appeared in the retrieved set (retriever configurations only). 1.00 = every gold article retrieved; lower values mark a retrieval miss — usually a multi-source question.

Question	survivor_6_1	survivor_final__grounding_surface
GT01	100.0%	100.0%

Question	survivor_6_1	survivor_final__grounding_surface
GT02	100.0%	100.0%
GT03	100.0%	100.0%
GT04	100.0%	100.0%
GT05	100.0%	100.0%
GT06	100.0%	100.0%
GT07	100.0%	100.0%
GT08	100.0%	100.0%
GT09	100.0%	100.0%
GT10	100.0%	100.0%
GT11	100.0%	100.0%
GT12	100.0%	100.0%
GT13	100.0%	100.0%
GT14	100.0%	100.0%
GT15	100.0%	100.0%
GT16	100.0%	100.0%
GT17	100.0%	100.0%
GT18	100.0%	100.0%
GT19	100.0%	100.0%

Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.



Task family	Questions	Baseline quality	Attention	Fail	Oracle quality	Oracle gap	Article recall
Exhaustive list or time filter	5	0.500	50.0	40.0%	n/a	n/a	100.0%
Cross-table quantitative reasoning	2	0.667	33.3	0.0%	n/a	n/a	100.0%
Evidence selection or explanation	3	0.778	22.2	11.1%	n/a	n/a	100.0%
Entity relationship or historical synthesis	6	0.972	2.8	0.0%	n/a	n/a	100.0%
Direct fact or table lookup	3	1.000	0.0	0.0%	n/a	n/a	100.0%

Baseline quality and failure rate use survivor_6_1. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.

Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

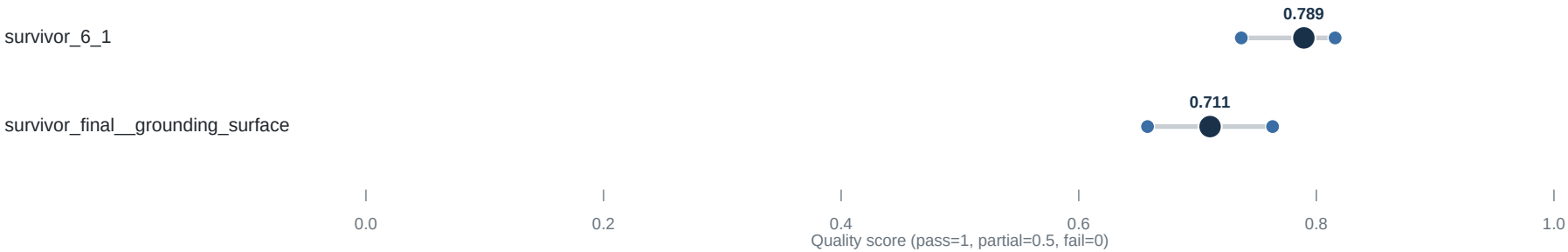
	survivor_6_1	survivor_final__grounding_surface
1 article	**0.803**	**0.727**
2 articles	**0.833**	**0.700**
3+ articles	**0.667**	**0.667**

Source scope	Configuration	Questions	Question-runs	Pass	Partial	Fail	Quality	Article recall
1 article	survivor_6_1	11	33	72.7%	15.2%	12.1%	0.803	100.0%
2 articles	survivor_6_1	5	15	66.7%	33.3%	0.0%	0.833	100.0%
3+ articles	survivor_6_1	3	9	66.7%	0.0%	33.3%	0.667	100.0%
1 article	survivor_final__grounding_surface	11	33	69.7%	6.1%	24.2%	0.727	100.0%
2 articles	survivor_final__grounding_surface	5	15	53.3%	33.3%	13.3%	0.700	100.0%
3+ articles	survivor_final__grounding_surface	3	9	66.7%	0.0%	33.3%	0.667	100.0%

Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.

Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.



Run	Configuration	Cases	Pass	Partial	Fail	Quality	Article recall	Full source coverage
eager_torvalds	survivor_6_1	19	12	4	3	0.737	100.0%	100.0%
hardcore_page	survivor_6_1	19	14	3	2	0.816	100.0%	100.0%
confident_stallman	survivor_6_1	19	14	3	2	0.816	100.0%	100.0%
admiring_diffie	survivor_final__grounding_surface	19	11	3	5	0.658	100.0%	100.0%
eloquent_hoare	survivor_final__grounding_surface	19	13	1	5	0.711	100.0%	100.0%
confident_zuckerberg	survivor_final__grounding_surface	19	13	3	3	0.763	100.0%	100.0%

Methodology and limitations

Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

Configuration grouping

Runs are grouped by a CSV experiment_name or config_name when one is present. If no explicit name is present, grouping falls back to evaluation mode, context source, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

Included experiment files

Run file
experiments/admiring_diffie.csv
experiments/confident_stallman.csv
experiments/confident_zuckerberg.csv
experiments/eager_torvalds.csv
experiments/eloquent_hoare.csv
experiments/hardcore_page.csv

Excluded files

Run file	Reason
datasets/wikipedia_GT_cost_inputs.csv	missing columns: ['score']
telemetry/admiring_diffie__llm_calls.csv	telemetry, merged into its run

Run file	Reason
telemetry/admiring_diffie__node_timings.csv	telemetry, merged into its run
telemetry/admiring_diffie__question_telemetry.csv	telemetry, merged into its run
telemetry/confident_stallman__llm_calls.csv	telemetry, merged into its run
telemetry/confident_stallman__node_timings.csv	telemetry, merged into its run
telemetry/confident_stallman__question_telemetry.csv	telemetry, merged into its run
telemetry/confident_zuckerberg__llm_calls.csv	telemetry, merged into its run
telemetry/confident_zuckerberg__node_timings.csv	telemetry, merged into its run
telemetry/confident_zuckerberg__question_telemetry.csv	telemetry, merged into its run
telemetry/eager_torvalds__llm_calls.csv	telemetry, merged into its run
telemetry/eager_torvalds__node_timings.csv	telemetry, merged into its run
telemetry/eager_torvalds__question_telemetry.csv	telemetry, merged into its run
telemetry/eloquent_hoare__llm_calls.csv	telemetry, merged into its run
telemetry/eloquent_hoare__node_timings.csv	telemetry, merged into its run
telemetry/eloquent_hoare__question_telemetry.csv	telemetry, merged into its run
telemetry/hardcore_page__llm_calls.csv	telemetry, merged into its run
telemetry/hardcore_page__node_timings.csv	telemetry, merged into its run
telemetry/hardcore_page__question_telemetry.csv	telemetry, merged into its run